Bangladesh University of Professionals

Department of Computer Science and Engineering (CSE)

Lab Report

Data Flow Diagram (DFD)

Course Title:

CSE 3206: Software Engineering LAB

Project Name:

Khet-Khamar

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Data Flow Diagrams (DFD)

Data Flow Diagrams (DFDs) are graphical representations used to illustrate how data moves through a system. In the Khet-Khamar platform, the DFDs demonstrate the interaction between users, processes, databases, and external services. These diagrams provide a structured overview of system functionality and data management.

DFD Level 0: Context Diagram

The Level 0 DFD, also known as the Context Diagram, presents the Khet-Khamar platform as a single unified system. It highlights the interaction between the system and external entities such as Guest Users, Registered Users, Administrators, and external APIs including Weather and News services.

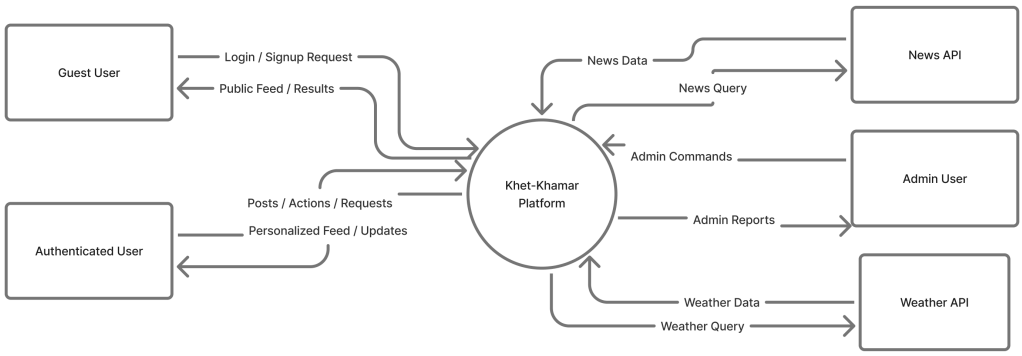

Figure 1: DFD Level 0: Context Diagram of the Khet-Khamar System

DFD Level 1: Process Decomposition

The Level 1 DFD expands the primary system into multiple functional modules. This decomposition provides a more detailed representation of internal processes including Authentication, Feed Management, Marketplace Operations, User Settings, and Database Interactions.

The diagram also demonstrates how information flows between processes and associated data stores, ensuring a clear understanding of system behavior and data handling.

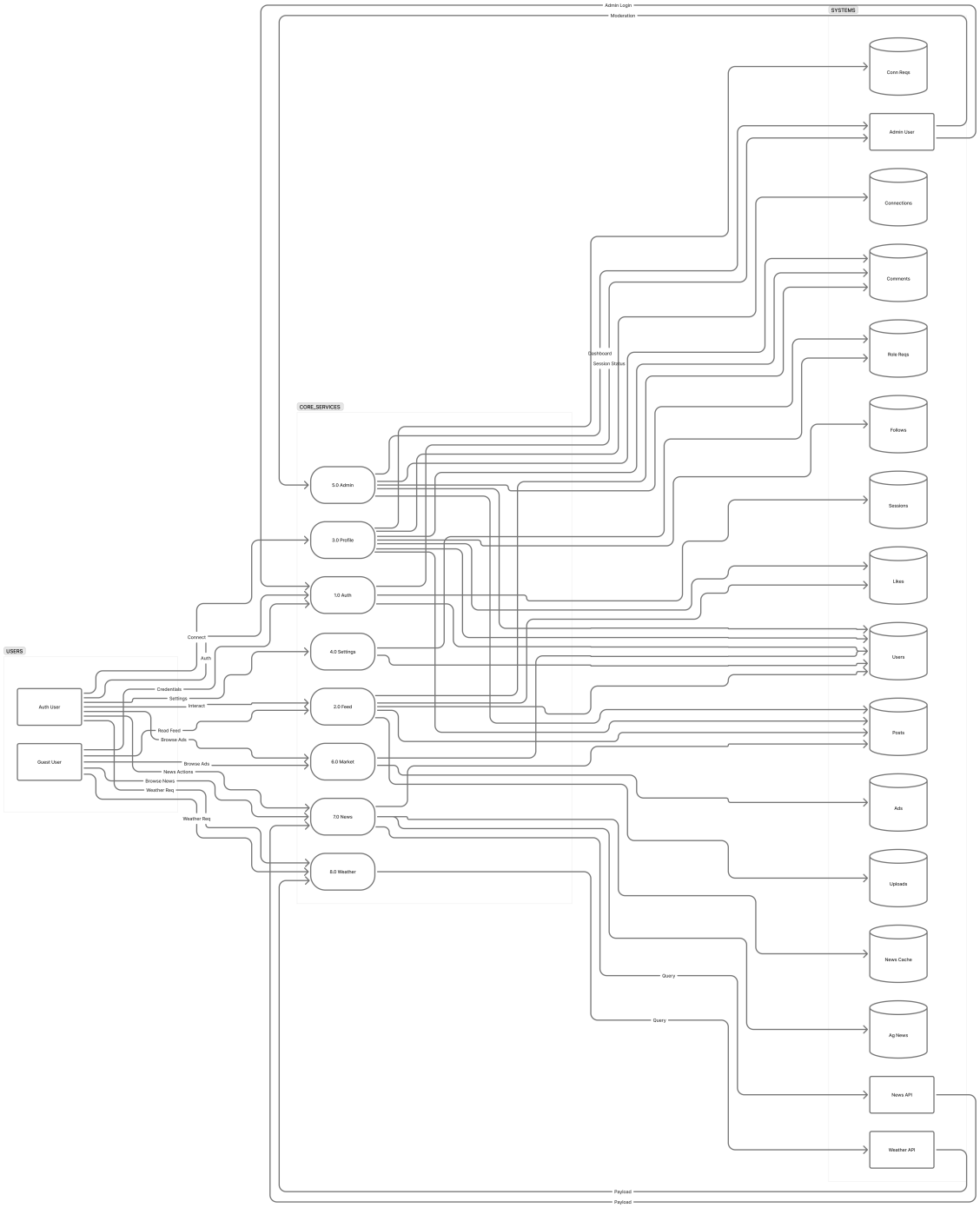

Figure 2: DFD Level 1: Functional Decomposition of the Khet-Khamar System

Balancing and Validation

To maintain consistency and correctness between the diagrams, the following balancing principles were followed:

- All external entities from the Context Diagram are preserved in the Level 1 DFD.

- User and administrator operations are separated to ensure better system clarity and security management.
- Data stores are accessed only through valid system processes, maintaining proper DFD standards.
- Information flow between processes remains logically consistent across all decomposition levels.

Conclusion

The presented Data Flow Diagrams provide a structured overview of the Khet-Khamar platform architecture and operational workflow. The Context Diagram offers a high-level understanding of external interactions, while the Level 1 DFD provides detailed insight into the internal system processes and data management structure. Together, these diagrams help ensure clarity, maintainability, and proper system documentation.